

University of the South Pacific

School of Information Technology, Engineering, Mathematics, and Physics

IS328 - Data Mining

Short Test Solutions

Semester 2, 2023

Possible Marks:	30
Weight:	5%
Time Allowed:	50 minutes
Number of Pages:	10 (including this page)

Learning Outcomes:

1. Explain the concepts, definitions and processes of Data Mining

Instructions:

1. Answer ALL questions.
2. Write ALL the answers in the spaces provided in the Answer Booklet.
3. If you require extra space for your answers, use the last page.

Q1: Pre-Processing (20 marks)

- a) What is data mining, and how does it differ from traditional data analysis methods? [2 marks]

Data mining is the process of discovering patterns, relationships, and insights from large datasets using techniques from various fields such as statistics, machine learning, and database management. Unlike traditional data analysis, which typically focuses on querying and summarizing data, data mining involves uncovering hidden information and knowledge that may not be immediately apparent.

- b) Describe the process of data preprocessing in data mining. Why is it important for the success of data mining tasks? [3 marks]

Data preprocessing is the step in data mining where raw data is cleaned, transformed, and prepared for analysis. It involves tasks like- data Cleaning, Data Discretization, data integration, Data transformation and data reduction - removing duplicate records, handling missing values, normalizing data, and reducing noise.

Proper data preprocessing is essential because it ensures the accuracy and reliability of the results obtained from data mining tasks

- c) What is the purpose of data reduction? Briefly describe two data reduction techniques.

[2 marks]

Data reduction is the process of minimizing the amount of data that needs to be mined or analyzed. Data reduction can increase computing efficiency and reduce complexity.

Two common techniques are sampling and feature selection

- d) Consider the following data (in increasing order) for the attribute size:

4, 5, 8, 15, 21, 21, 24, 25, 25, 28, 34, 36

Use smoothing by bin boundaries to smooth these data, using a bin depth of 3. [2 marks]

Bins (depth 3)

Bin 1: 4, 5, 8

Bin 2: 15, 21, 21

Bin 3: 24, 25, 25

Bin 4: 28, 34, 36

After smoothing by boundaries

Bin 1: 4, 4, 8

Bin 2: 15, 21, 21

Bin 3: 24, 25, 25

Bin 4: 28, 36, 36

e) For the data set 5, 6, 12, 13, 15, 18, 22, 50, find the following:

(i) Normalize the above data set by min-max normalization. [4 marks]

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

$$v = \frac{5 - 5}{50 - 5} (1 - 0) + 0 = 0 \quad v = \frac{6 - 5}{50 - 5} (1 - 0) + 0 = 0.022$$

$$v = \frac{12 - 5}{50 - 5} (1 - 0) + 0 = 0.156 \quad v = \frac{13 - 5}{50 - 5} (1 - 0) + 0 = 0.178$$

$$v = \frac{15 - 5}{50 - 5} (1 - 0) + 0 = 0.22 \quad v = \frac{18 - 5}{50 - 5} (1 - 0) + 0 = 0.29$$

$$v = \frac{22 - 5}{50 - 5} (1 - 0) + 0 = 0.38 \quad v = \frac{50 - 5}{50 - 5} (1 - 0) + 0 = 1$$

(ii) In z-score normalization, what value should the first number 5 be transformed to? [2 marks]

Step 1: Calculate the mean (μ)

$$\mu = \frac{5+6+12+13+15+18+22+50}{8}$$

$$\mu = \frac{141}{8}$$

$$\mu = 17.625$$

Step 2: Calculate the variance and standard deviation (σ)

To find the variance, we need the average of the squared differences from the mean:

Variance =

$$\frac{(5-17.625)^2 + (6-17.625)^2 + (12-17.625)^2 + (13-17.625)^2 + (15-17.625)^2 + (18-17.625)^2 + (22-17.625)^2 + (50-17.625)^2}{8}$$

Breaking it down:

$$= \frac{159.390625 + 134.640625 + 31.515625 + 21.390625 + 6.890625 + 0.140625 + 19.140625 + 1048.140625}{8}$$

$$= \frac{1421.15625}{8}$$

$$= 177.64453125$$

Now, the standard deviation (σ) is the square root of the variance:

$$\sigma = \sqrt{177.64453125}$$

$$\sigma \approx 13.33 \text{ (rounded to two decimal places)}$$

Step 3: Calculate the z-score for the number 5

Using the z-score formula:

$$z = \frac{x - \mu}{\sigma}$$

For $x = 5$:

$$z = \frac{5 - 17.625}{13.33}$$

$$z \approx -0.946 \text{ (rounded to three decimal places)}$$

Answer:

Using z-score normalization, the first number (5) in the dataset would be transformed to approximately -0.946.

- f) A sample data table containing attributes of mixed type is shown below. Determine the attribute type for attributes *Attribute-1*, *Attribute-2* and *Attribute-3*. [3 marks]

Object	Attribute-1	Attribute-2	Attribute-3
1	Yes	Small	44
2	No	Medium	20
3	No	Large	21
4	Yes	Small	15

Attribute-1 = Asymmetric binary
Attribute-2 = Ordinal
Attribute-3 = numeric

- g) Consider the two relations R and S from two different databases to be integrated to obtain a single table Restaurant. Both relations contain records that describe Restaurant entities in the real-world. Carry out schema integration by resolving semantic conflicts (only) to map the above tables to a single Restaurant table. You don't have to carry out the data (instance) integration. [2 marks]

Table R

Name	Street	City	Cuisine	Supervisor	Operating Since
VillageWok	Laucala Bay	Suva	Chinese	Hing	1970
Ching	Matintar	Nadi	Chinese	Libby	1990
OldCountry	Gordon St	Suva	Italian	Tom	2000

Table S

Restaurant Name	Location City	Manager	Years Old	Specialty
VillageWok	Suva	Hing	40	Chinese
Ching	Nadi	Libby	20	Chinese
OldCountry	Suva	Tom	10	Italian

Name	City	Cuisine	Supervisor	Operating Since
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Q2: Classification Techniques (10 marks)

- a) Consider the following decision tree in figure 1 below to answer the following questions:

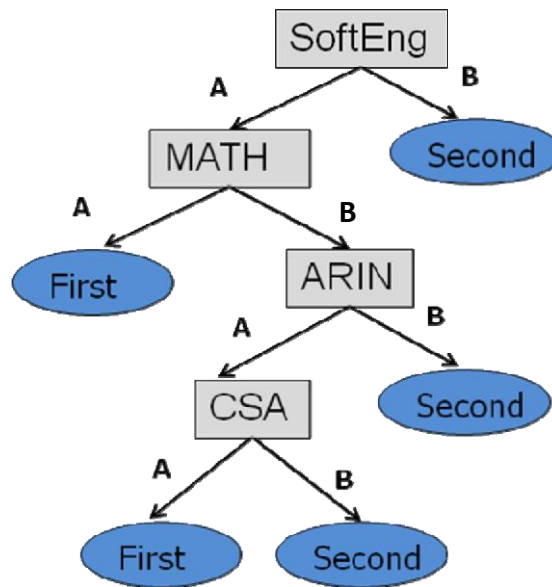


Figure 1

- i) Derive all possible rules from the decision tree and write them as IF-THEN rules. [2.5 marks]

IF SoftEng = B THEN Second

IF SoftEng = A AND Math = A THEN First

IF SoftEng = A AND Math = B AND Arin = B THEN Second

IF SoftEng = A AND Math = B AND Arin = A AND CSA = A THEN First

IF SoftEng = A AND Math = B AND Arin = A AND CSA = B THEN Second

- ii) Use the decision tree in Figure 1 to briefly explain the concept behind the classification technique. [1.5 marks]

The concept behind classification is to use a set of rules or a model, like the decision tree, to classify, according to the values of the attributes, previously unseen examples to a particular class. In Figure 1, the students are classed as either First or Second according to the decision tree.

- iii) Discuss the concept of entropy in decision tree algorithms. How is entropy used to determine the best attribute for splitting? [2 marks]

Entropy is a measure of impurity or disorder in a dataset. In decision trees, entropy is used to evaluate how well an attribute splits the data into homogeneous classes. The attribute that results in the most significant reduction in entropy after splitting is chosen as the best attribute for the next node. Lower entropy indicates better separation and more informative splits.

- b) Use the confusion matrix below to answer the questions that follow. [4 marks]

	Birds	Mammals
Birds	10	2
Mammals	3	15

- i) What percent of the mammal instances were incorrectly classified?

Total actual mammal instances = 3 (incorrectly classified as Birds) + 15 (correctly classified as Mammals) = 18

Incorrectly classified mammal instances = 3

So, the answer is $3/18 \times 100\% = 16.67\%$

Answer = $3/18 = 16.67\%$ – 1 Mark

- ii) What percent of the bird instances were correctly classified?

Total actual bird instances = 10 (correctly classified as Birds) + 2 (incorrectly classified as Mammals) = 12

Correctly classified bird instances = 10

So, the answer is $10/12 \times 100\% = 83.33\%$

Answer = $10/12 = 83.33\%$ – 1 Mark

iii) What is the accuracy and error rate of this classification model?

- Total instances = $10 + 2 + 3 + 15 = 30$
- Correctly classified instances (True Positives) = 10 (Birds correctly classified as Birds) + 15 (Mammals correctly classified as Mammals) = 25
- Incorrectly classified instances (False Positives + False Negatives) = $2 + 3 = 5$

$$\text{Accuracy} = \frac{\text{Correctly classified instances}}{\text{Total instances}} = \frac{25}{30} \times 100\% = 83.33\%. \text{ This answer is correct.}$$

$$\text{Misclassification Rate (or Error Rate)} = \frac{\text{Incorrectly classified instances}}{\text{Total instances}} = \frac{5}{30} \times 100\% =$$

Accuracy = $25/30 = 83.33\%$ – 1 Mark

Misclassification Rate = $5/30 = 16.67\%$ – 1 Mark

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